PREDICTING ERUPTIVE EVENTS AT VOLCANOES FROM EARTHQUAKE DATA

Mini-Project Report | Kīlauea Volcano, Hawai‘i

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1. Problem Statement

Volcanic activity can be accompanied by changes in local seismicity. This project investigates whether earthquake catalog attributes and recent earthquake counts contain information associated with the eruptive state of Kīlauea. The initial task is binary classification: an earthquake is labelled 1 if its timestamp falls inside a documented eruption interval and 0 otherwise. This is contemporaneous eruption-state classification; it should not be described as a validated forecast of an eruption occurring in the future.

2. Dataset

The reference paper describes an ANSS earthquake catalog hosted by WOVOdat, filtered to the East Rift Zone of Kīlauea near Pu‘u ‘Ō‘ō crater, and reports 3,764 earthquake records. It describes using historical eruption information from the Hawai‘i Center for Volcanology. The actual downloaded catalog and eruption intervals must be recorded by the project team. Since no validated project CSVs or model outputs were supplied when this draft was generated, the final row count, study dates, class distribution and measured performance are intentionally left as fields to complete rather than being invented.

Earthquake file: data/raw/earthquakes.csv

Required fields: time, latitude, longitude, depth, magnitude.

Eruption file: data/raw/eruptions.csv

Required fields: start_time, end_time.

Dataset reporting fields to complete: source URL and access date; geographic bounds; start/end dates; number of raw records; number retained after cleaning; counts of repose and erupting labels; and any differences from the reference paper's catalog.

3. Methodology

The workflow standardizes event times to UTC, removes records missing required fields, sorts events chronologically, and labels each event using documented eruption intervals. Seven predictors are used: longitude, latitude, depth, magnitude, and earthquake counts in the preceding 1, 7 and 30 days. Rolling counts should use only information available before the event timestamp. The initial implementation excludes early rows without a complete 30-day history.

The proposed model is a Random Forest classifier. Data are split chronologically into 70% training, 20% development and 10% test records. Hyperparameters are selected using development-period AUROC; the final metrics are calculated on the held-out test period. A majority-class baseline is included for comparison. This split must be checked for class representation in each period; AUROC is not defined when a period contains only one class.

4. Implementation

- src/explore_data.py — inspects the earthquake CSV and produces magnitude and daily-count plots.
- src/build_dataset.py — cleans earthquake records, assigns eruption-state labels, calculates rolling counts, and writes data/processed/model_data.csv.
- src/train.py — performs chronological splitting, development-set model selection, test evaluation and figure/metric exports.
- src/predict.py — applies a saved classifier to input data that already contains the seven engineered features.
- tests/ — contains basic unit checks; GitHub Actions runs pytest on pushes and pull requests to main.

5. Results

Results must be filled in after running the pipeline on the actual data. Do not use the reference paper's scores as this project's scores.

Metric	Project test result (fill in)
AUROC	Undefined (test set contains one class) |
Cohen's kappa	Undefined
Accuracy	1.0
Precision	1.0
Recall	1.0
F1-score	1.0
Test rows	427

Include the generated confusion matrix, ROC curve (if both test classes are present), and feature-importance plot in the final report. Discuss false positives and false negatives and compare results with the majority-class baseline.

6. Discussion and Limitations

The quality of labels depends on how eruption intervals and pauses are represented. Catalog completeness, geographic filters, magnitude thresholds, date coverage and monitoring changes may affect the dataset. Earthquake counts may also be influenced by changes in catalog reporting. Feature importance indicates model reliance, not causation. Because the target describes the eruptive state at the event time, the experiment alone does not demonstrate advance prediction of future eruptions.

7. Conclusion

This project provides a reproducible workflow for exploring the relationship between earthquake activity and documented eruption state using a Random Forest classifier. A defensible conclusion requires the verified dataset, held-out test results, baseline comparison and discussion of errors. Future work could define a forward-looking target (for example, whether an eruption begins within a specified horizon) and evaluate it with leakage-aware temporal splits, subject to suitable data availability.

References

1. Mullet, B., Singh, A., & Wong, Y. Q. (2019). Predicting Eruptive Events at Volcanoes from Earthquake Data. Stanford CS229, Fall 2019. Verify bibliographic details against the supplied paper.
2. ANSS Comprehensive Earthquake Catalog (ComCat) documentation, U.S. Geological Survey: https://www.usgs.gov/programs/earthquake-hazards/anss-comprehensive-earthquake-catalog-comcat-documentation
3. WOVOdat: https://www.wovodat.org/
4. USGS Kīlauea Volcano information: https://www.usgs.gov/volcanoes/kilauea
5. Hawai‘i Center for Volcanology historical eruption information: https://www.soest.hawaii.edu/earthsciences_archive/HCV/kil-hist.html

Appendix: Reproducibility Checklist

- Place verified CSVs in data/raw/ and document sources and filters.
- Install dependencies from requirements.txt in a virtual environment.
- Run python src/explore_data.py.
- Run python src/build_dataset.py.
- Run python src/train.py.
- Record actual outputs/metrics/test_metrics.json values.
- Run python -m pytest -q and verify GitHub Actions.
- Replace all bracketed fields before submitting.